

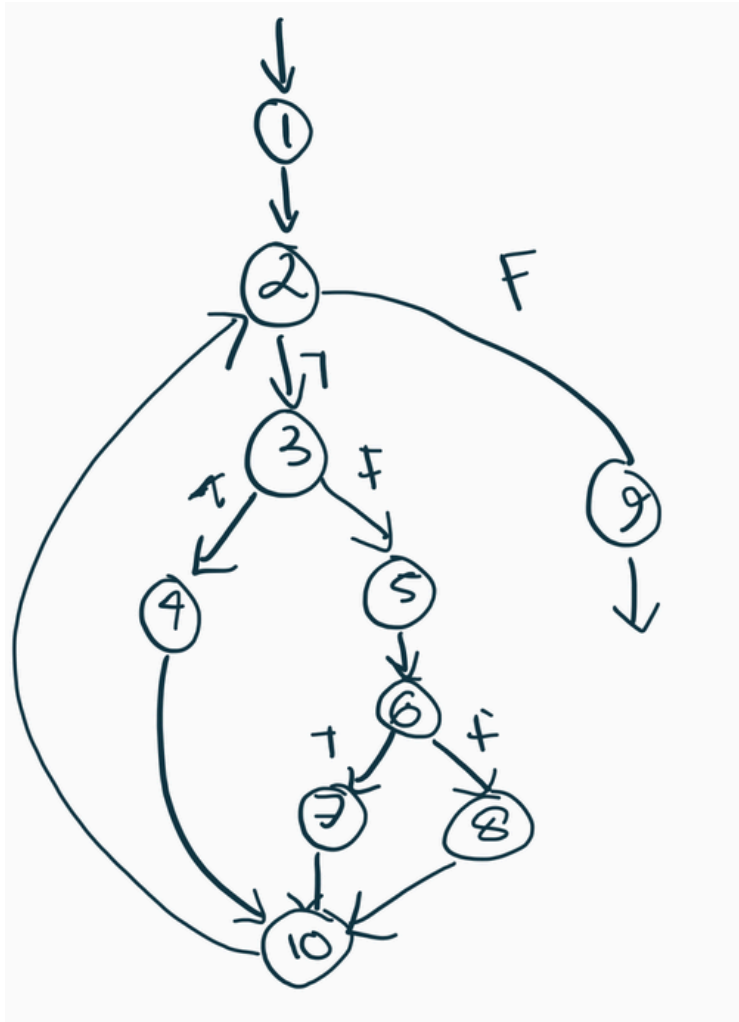
CFG and Cyclomatic Complexity

Given this code, answer the following questions:

Python	Java
<pre>def process_numbers(nums): result = [] for num in nums: if num % 2 == 0: continue else: result.append(f'{num} is odd') if num > 10: result.append(f'{num} is greater than 10') else: result.append(f'{num} is 10 or less') return result</pre>	<pre>public List<String> processNumbers(List<Integer> nums) { List<String> result = new ArrayList<>(); for (int num : nums) { if (num % 2 == 0) { continue; } else { result.add(num + " is odd"); } if (num > 10) { result.add(num + " is greater than 10"); } else { result.add(num + " is 10 or less"); } } return result; }</pre>

- Draw the CFG for the given method
- Calculate the Cyclomatic Complexity number for the given method, and find the Basic Paths Set
- Generate test cases for all paths in Basic Paths Set

Answer:



b.

Cyclomatic Complexity Number = $P + 1 = 3 + 1 = 4$

Basic Paths Set =

[1-2-3-4-10-2-3-5-6-7-10
-2-3-5-6-8-10-2-9]

Test Cases:

nums = [2, 11, 9]

```
def print_primes(n):  
    for i in range(2, n + 1):  
        is_prime = True  
        for j in range(2, int(i ** 0.5) + 1):  
            if i % j == 0:  
                is_prime = False  
                break  
        if is_prime:  
            print(i)
```